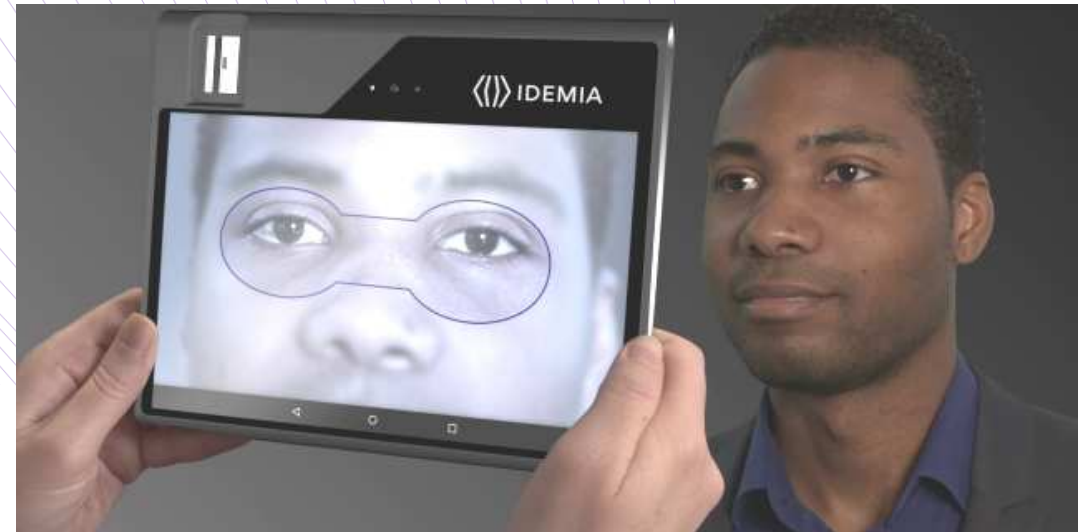


MorphoTablet 2i

Overview / Introduction

Safety Precautions

Best Practices Guide For Better Iris Capture



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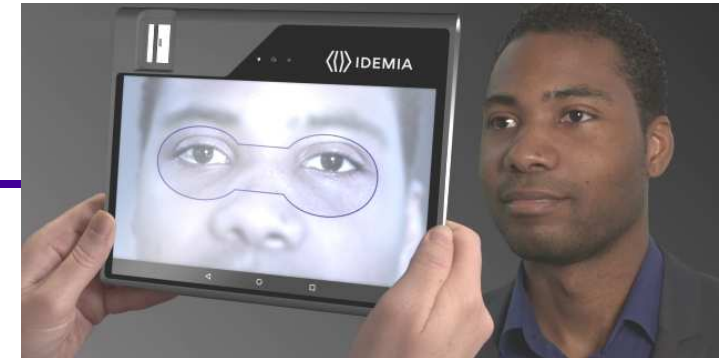
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MorphoTablet 2i Capture Feature Overview





MorphoTablet 2i Capture Feature Introduction



- MT2i is a biometric mobile Android device which captures the two irises simultaneously. The Tablet is dedicated for operated usage, and not self acquisition.
 - The product can detect irises between 15cm and 35cm of distance. For best results keep the eyes between 20-30 cm (~8-12 inches) away for optimum recognition.
 - The performance of the iris recognition is correlated to the quality of the captured iris image based on the operator use and the cooperation of the user. If the capture process is carried out properly by a trained operator following the best practices, the images will be suitable for iris recognition purposes.
- **Caution! Please follow all instructions before using the iris Capture feature**

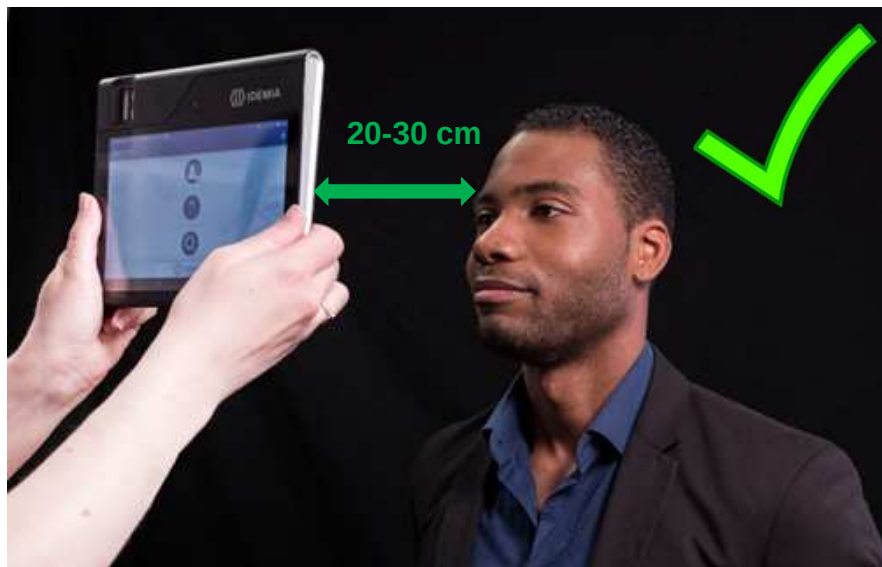


Iris Capture Safety Precautions



Caution! Please follow all instructions before using the iris capture feature

- The operator shall inform the user to NOT stare at the iris recognition LED light when its eyes are less than 15 cm (~6 inches) away from the device.
- The operator must be at the right distance and must keep a distance between 20 - 30cm between the device and the person.
- Do not use iris recognition with infants. Doing so may damage eyesight
- Prolonged use can cause eye-strain
- The device will automatically stop iris capture after 10 seconds.

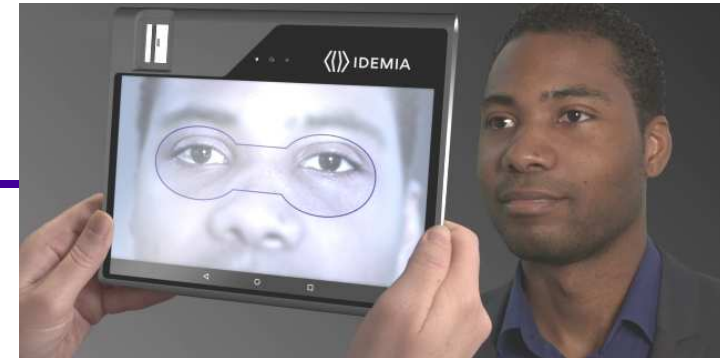


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Step two: CAUTION !



Iris Capture Basics



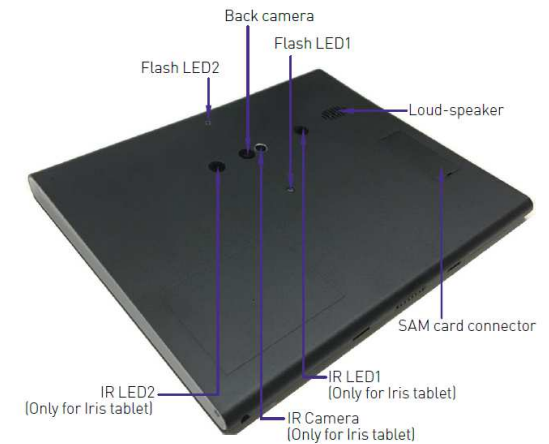
For Better Iris Capture

- The operator must ask the person to keep its eyes open (avoid occlusions = eyelid covers the iris, avoid narrowed eyes) during the capture
- The operator must verify that the person does not wear glasses
- Marketing video is a good support for enrolment and identification.





Iris Capture Basics



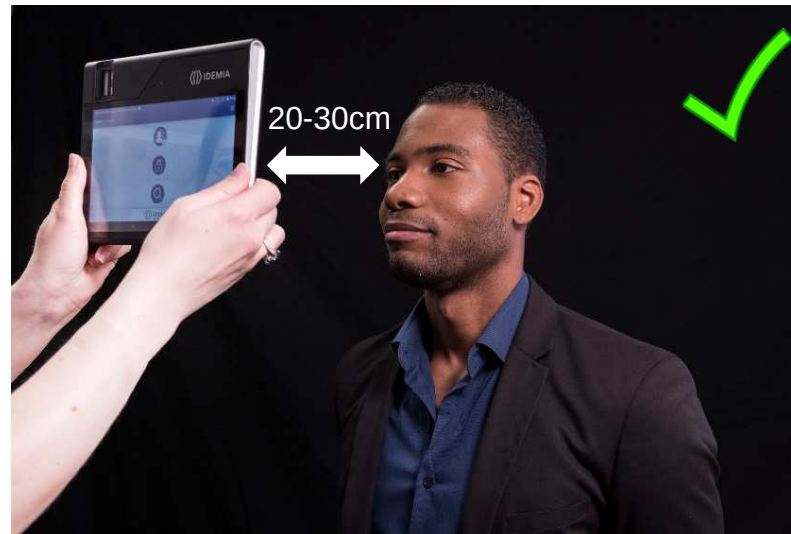
- If the area around the Iris Camera, Iris LEDs, is dirty, wipe it with an antistatic, soft, dry cloth before using the iris capture feature.
- Do not expose the device to excessive impacts or scratch the area around the Iris Camera and LED with sharp or hard objects; this may damage the LED or camera, causing it to malfunction.
- The device may not perform properly if you apply stickers, ink or paint on or around the Iris Camera or LED.
- The device may not perform properly in places with strong light, reflected light.



Operator Placement

For Better Iris Capture

- The operator must be at the same height as the person (seated if seated, standing if standing) to have the tablet in front and in the axis of the eyes (tablet and head straight), and avoid angle.
- The operator must avoid area which may hinder mobility, flexibility during Iris capture: such as table, ...



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Operator Placement

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- The operator must place the tablet in front of the person's eyes before starting the acquisition. The arms of the operator should not move up and down (to bring the tablet in front of the person's eyes) before starting the acquisition.
- To ease the operator to be at the right distance, user indications (Put eyes in clear area, stay here, closer, farther) are displayed on the top of the screen over a distance 15 to 35cm of the person's eyes
 - Put eyes in clear area: Iris is not found. User is not well positioned
 - Stay here: Iris is found at the expected radius. Operator should hold tablet still and user should hold his head still to get best iris capture results
 - Closer: Iris is found but radius of it is too small. Operator has to bring the tablet closer to user.
 - Farther: Iris is found, but iris is bigger then the expected radius. Operator has to get the tablet farther from user's face to avoid to get some blur on the found iris image



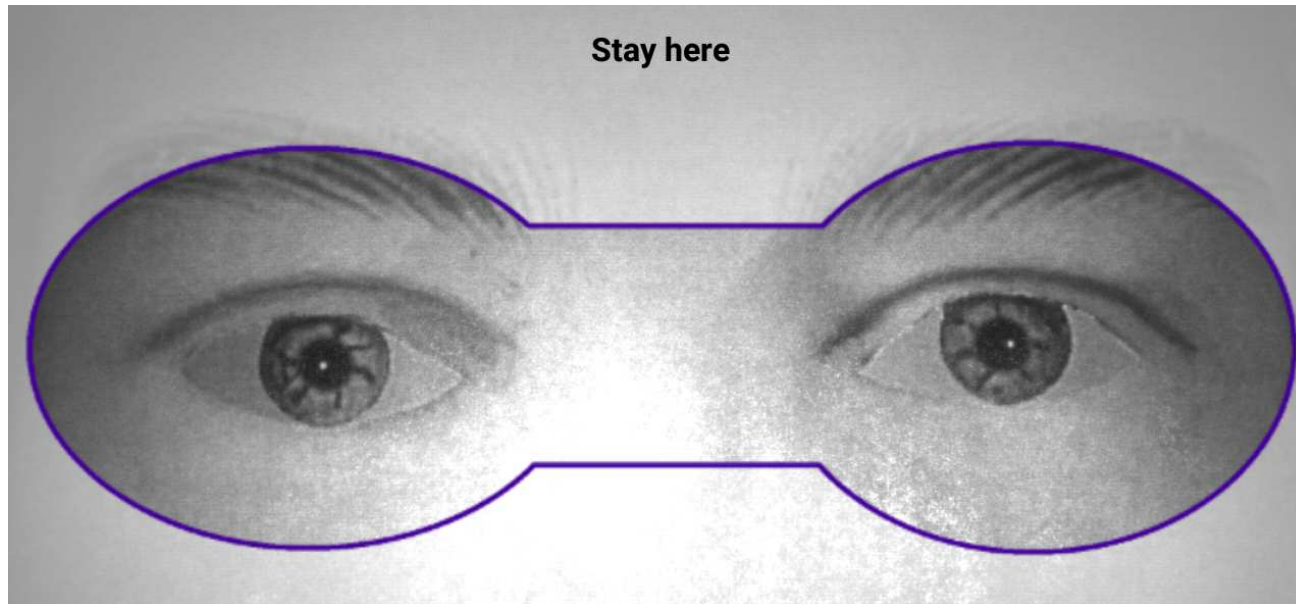
Enrolment

- **Recommendation**

- Follow the Operator Placement section.
- Launch acquisition activity by clicking of “Enroll” Button.
- The video is displayed on the screen.

- It is necessary to stay (stable/do not move) to keep “stay here” message as much as possible (avoid “closer”, “farther”)
- Iris is captured when message on the top of the screen such as “stay here” disappear

The screenshot shows the 'Iris sensor' software interface with tabs for 'ENROLLMENT', 'IDENTIFICATION', and 'AUTHENTICATION'. The 'ENROLLMENT' tab is active. It features a 'Start Enrollment' label and an 'Enroll' button. Below this, there is a video feed area showing a white silhouette of a head against a teal background. To the right of the video, there are input fields for 'First Name' and 'Last Name', both with placeholder text 'enter your first name' and 'enter your last name' respectively. A 'Generate ID randomly!' button with a dice icon is also present. A 'Display Images' button is located below the name fields.



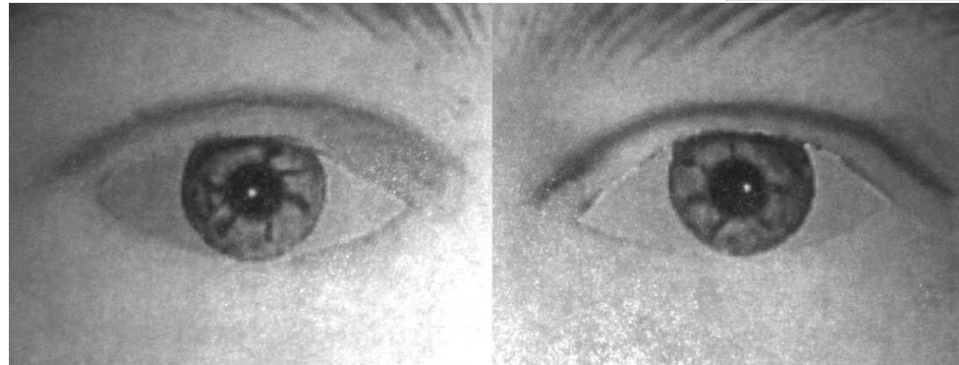
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Enrolment

- **Recommendation**
- When the acquisition is well performed, it does display the following screen:
- “Display Images” Button show the images acquired for the both eyes:

The screenshot shows the 'ENROLLMENT' tab of the 'Iris sensor' application. At the top, there are tabs for 'ENROLLMENT', 'IDENTIFICATION', and 'AUTHENTICATION'. Below the tabs, there is a 'Start Enrollment :' label and an 'Enroll' button. The main area displays a silhouette of a head with a white iris sensor icon. To the right of the silhouette, there are input fields for 'First Name: Charles' and 'Last Name: Robert'. Below these fields is a 'Display Images' button. To the right of the 'Display Images' button is a 'Generate ID randomly!' button with a dice icon. At the bottom right, there is an 'Enrollment OK' button.



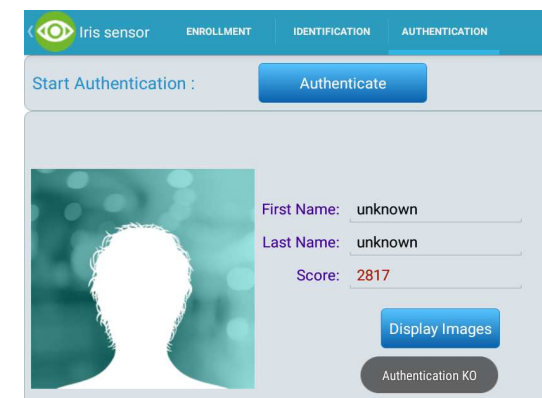
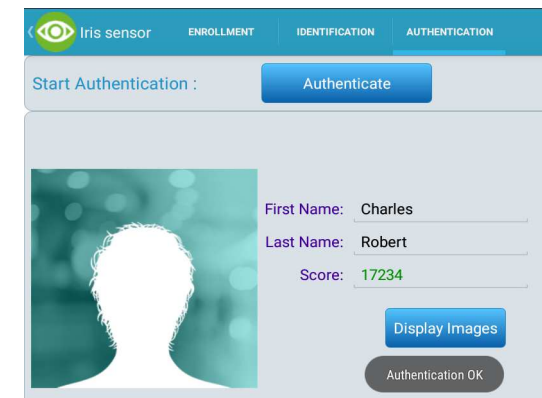
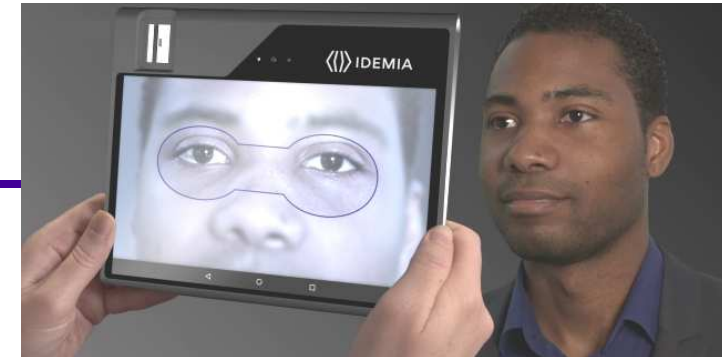
- If “Enrollment is KO” is displayed, then an eye has not been captured. It could be a light environment problem
 - (which makes the variation between the black and white of the eye is not large for the algorithm). Please move slightly.



Authentication

- **Recommendation**

- Follow the Operator Placement section
- Authentication process starts with same screen as enrollment.
 - The video is displayed on the screen.
- When user is recognized (and matching score is displayed) user details are display.:
- If user is not authenticated, the following error screen is displayed:.



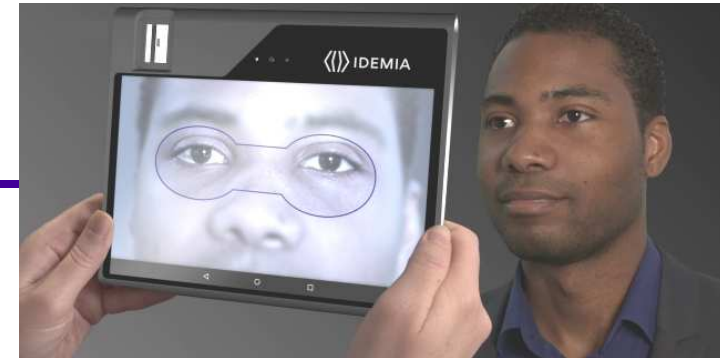


Identification

- **Recommendation**

- Follow the Operator Placement section
- Identification does act as authentication.

- The first user matching (score greater score than 4000) is returned as below:



 Iris sensor

ENROLLMENT IDENTIFICATION AUTHENTICATION

Start Identification : Identify



First Name:

Last Name:

Score:

Display Images

Identification OK

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Settings

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- **Recommendation**

- Acquisition Timeout :
 - After the Timeout, the device will automatically stop iris capture.
 - 10 sec. is the maximum recommended figure to set.



- Matching Threshold :
4 000 is recommended figure to set for demonstration purpose

 **General**

Acquisition Timeout (in Milliseconds)
10000

Matching Threshold (minimum Score)
4000

- It is possible to delete all records

 **Records**

BIOMETRIC DATABASE

Delete all records

 Iris sensor | ENROLLMENT | IDENTIFICATION | AUTHENTICATION

Start Enrollment : **Enroll**

First Name:  Generate ID randomly!

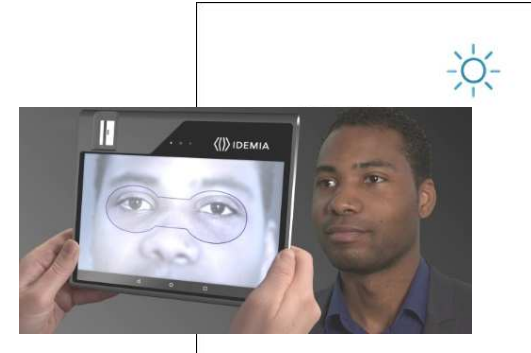
Last Name: **Display Images**



Iris Capture: Lighting Environment and Massive Enrolment

- **Outdoor Recommendation**

- No direct sunlight shall reach Iris Camera
- The operator (holding the tablet) and the person to be enrolled must be in the shade (under a yard)
- The operator (holding the tablet) must be back to the sun



Avoid Direct Sunlight
Avoid Direct Window

- **Indoor Recommendation**

- The operator (holding the tablet) must be back to the window
- The operator must avoid having the light (light from the desk, window, LED panels, ...) on the Iris part of the tablet (Iris Camera, Iris LEDs on the back of the tablet)
- Best Practice: The operator could check if the ceiling light emits in the IR → look through the IR tablet the video is displayed on the screen

- **Massive Enrolment**

- The operator must leave at least 70sec between 2 acquisitions with the same tablet to allow time for the IR leds to return to room temperature (a led that heats emits less).